

Title and Authors.

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FAIR, Meta; Northeastern University

01 Title and Authors.

- Present paper title and authorship details
 - Yuhan Shen^{1,2*} · Huiyu Wang¹ · Xitong Yang¹ · Matt Feiszli¹
 - Ehsan Elhamifar² · Lorenzo Torresani¹ · Effrosyni Mavroudi¹
 - 1 FAIR, Meta · 2 Northeastern University
- Affiliations highlight collaboration across institutions
 - FAIR, Meta for large-scale vision-language research
 - Northeastern University for academic partnership
 - Cross-lab synergy supports **strong results**
- Acknowledgment of equal contribution and roles
 - Lead authors coordinate experiments and writing
 - Senior authors advise methodology and positioning
 - Community resources enable **reproducible research**

02 Teaser: NVOS

1. pesto sauce jar

2. pasta

3. pan

4. pesto sauce

5. wooden spoon

6. hob/cooktop/stovetop

stir pasta



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stir pasta



Narration:
The person uses a [wooden spoon]<5> to stir [pasta]<2> in a [pan]<3> on the [stovetop]<6>.

- Segment narrated object instances without spatial supervision
 - Align noun phrases to mask proposals via weak supervision
 - ROSA framework for pixel-level grounding
- Predict phrase-specific masks for each video frame
 - Output multiple instance masks per narration
 - Support for overlapping and occluded objects
- Demonstrate competitive zero-shot performance
 - Large gains over weak baselines on benchmarks
 - Generalization beyond egocentric domain
 - Results indicate **robust transfer**

03 Motivation and NVOS

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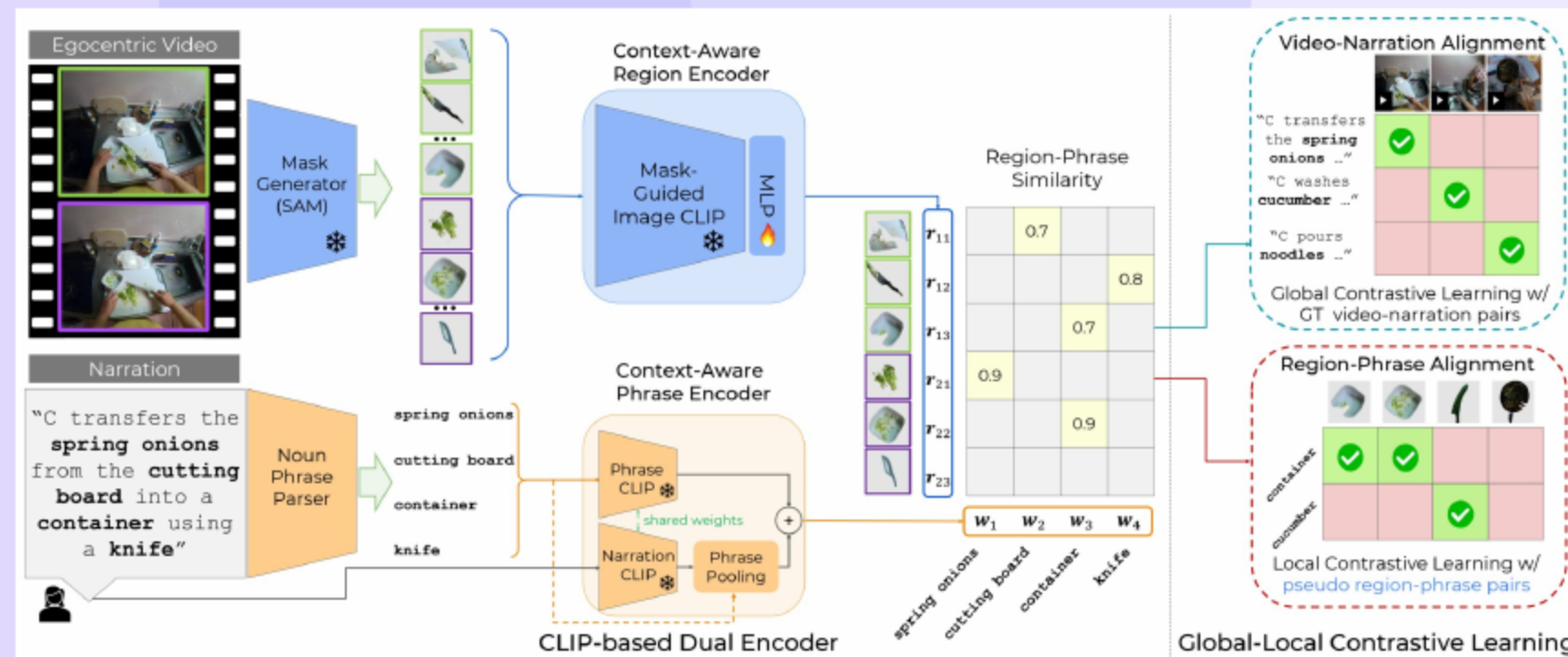
6. hob/cooktop/stovetop

stir pasta



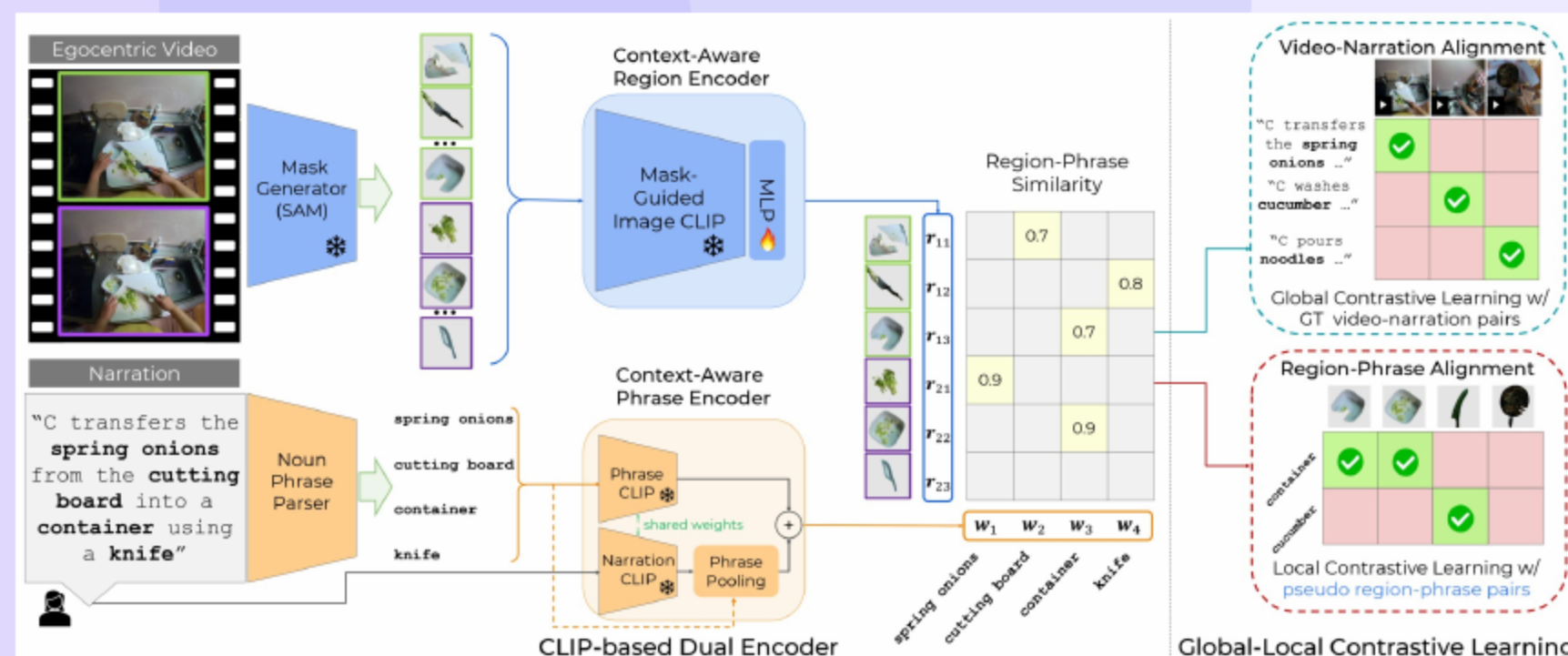
Narration:
The person uses a [wooden spoon]<5> to stir [pasta]<2> in a [pan]<3> on the [stovetop]<6>.

- NVOS addresses fine-grained egocentric grounding demands
 - Instance-level, phrase-specific masks per frame
 - Handles occlusion and overlap beyond boxes
 - Supports multi-object phrases in narration
- NVOS differs from referring expression segmentation
 - Goes beyond single-object focus
- NVOS advances over open-vocabulary segmentation
 - Moves from category labels to instances
 - Disambiguates nearby similar objects
 - Language supervision enables **finer control**

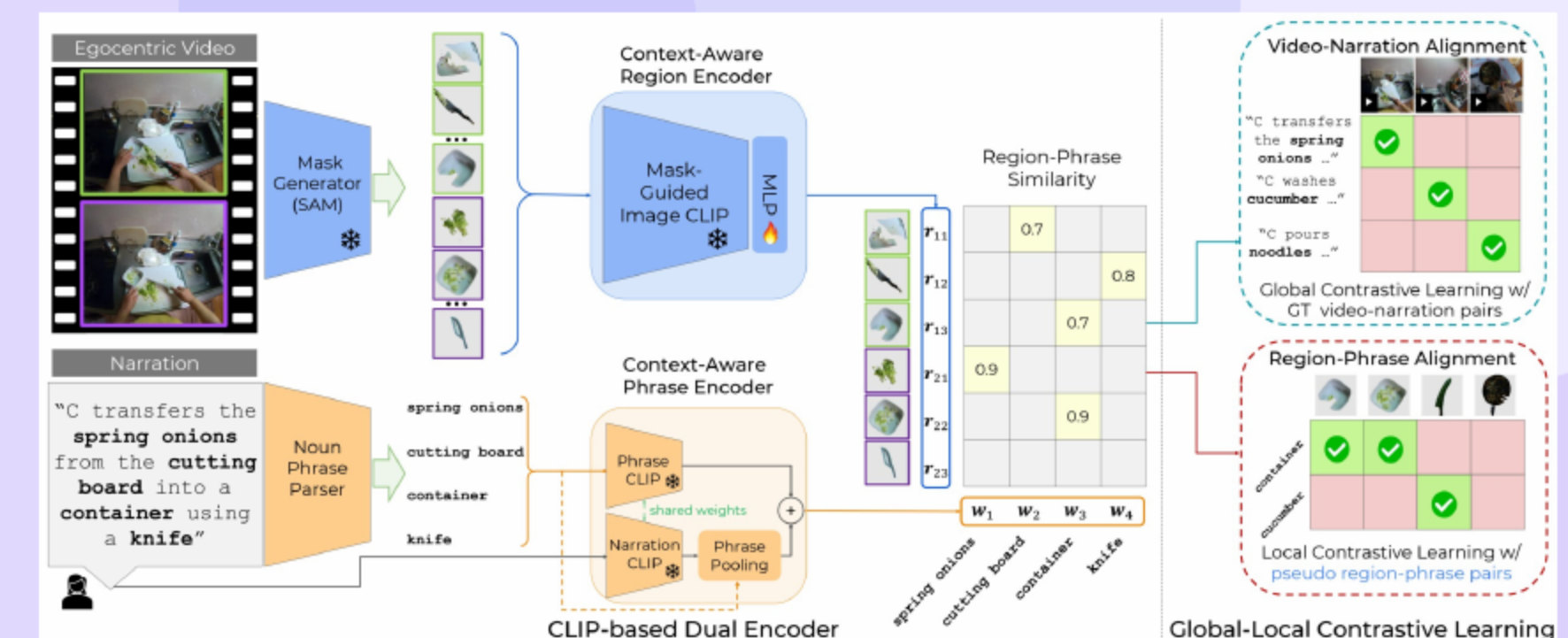


- ROSA performs weakly supervised region-phrase alignment
 - Use SAM proposals for class-agnostic masks
 - Learn similarities with CLIP priors
 - Produce **phrase-conditioned masks**
- Global video-narration score aggregates local matches
 - Temporal pooling emphasizes informative frames
- Contributions enable comprehensive zero-shot evaluation
 - VISOR-NVOS: 14,612 clips and 37,170 objects
 - State-of-the-art zero-shot on VISOR-NVOS and VOST
 - Transfer to YouCook2-BB shows **domain generalization**

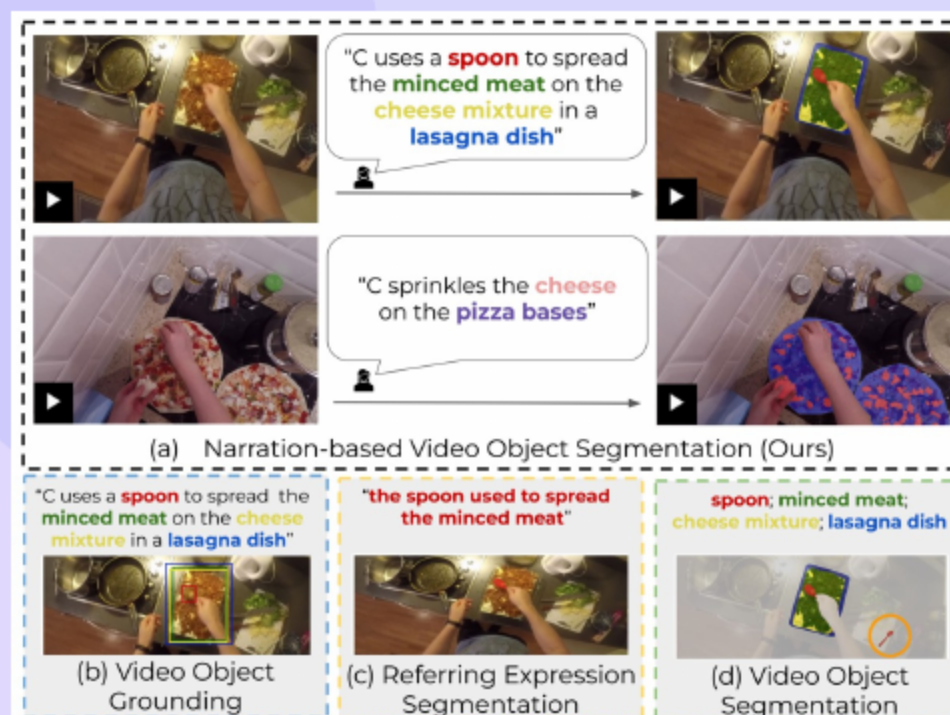
- Pixel-level egocentric grounding advances beyond coarse boxes
 - Transition from third-person to egocentric settings
 - Avoid reliance on mouse traces or points
 - Achieve **fine-grained alignment**
- Language-only supervision contrasts semi-supervised VOS
 - No first-frame masks required
 - Instance-specific rather than category-level
 - Broader applicability with **lower labeling cost**
- Global and local contrastive learning adapts CLIP to regions
 - Region and phrase embeddings for masks
 - Joint video-narration and region-phrase objectives
 - Improved **pixel-level grounding**



- Model inputs and outputs define phrase-conditioned segmentation
 - Input frames with narration noun phrases
 - Output masks per phrase and frame
 - Training uses **no spatial labels**
- SAM proposals provide class-agnostic regions for alignment
- Inference selects highest-scoring mask per phrase per frame
 - Compute region-phrase similarity
 - Temporal pooling informs video-level scores
 - Top mask yields **final segmentation**



- Mask-guided image encoder produces proposal embeddings
 - Share early computation and mask attention late
 - MLP projects to region features
 - Design preserves **scene context**
- Phrase encoder combines narration-aware and phrase-only signals
- Global and local objectives supervise alignment without labels
 - Video-narration contrastive objective over batch pairs
 - Region-phrase objectives with pseudo alignments
 - Combined loss strengthens **pixel grounding**



- VISOR-NVOS provides phrase-linked instance masks in egocentric clips
 - 14,612 clips and 37,170 referred objects
 - Validation and test splits without training overlap
 - Rich narrations support **fine supervision**
- VOST evaluates robustness under strong appearance changes
- Evaluation focuses on zero-shot generalization
 - Train only on Ego4D narrations
 - No spatial labels or fine-tuning
 - Benchmarks measure **mask quality**

- Training uses Ego4D narrations with short clips and sampled frames
 - 250k one-second clips with four frames
 - About two narrations per clip on average
 - Sampling strategy encourages **diverse coverage**
- Metrics evaluate region accuracy and boundary quality
 - VISOR-NVOS uses J, F, and J and F
 - VOST reports union and instance IoU
 - Comprehensive scores reflect **holistic quality**
- Implementation relies on SAM proposals and CLIP encoders
 - Grid prompts and non-maximum suppression
 - ViT-L or ViT-B backbones with MLP head
 - Loss weighting balances **global and local** learning

- Weakly supervised methods combine SAM with language models
 - SAM with CLIP similarity for scoring (Kirillov et al., 2023; Radford et al., 2021).
 - CoMMa with SAM prompting from points (Gao et al., 2023; Kirillov et al., 2023).
 - Serve as **language-only** references
- Stronger supervised methods leverage labeled detections or masks
 - ODISE for open-vocabulary segmentation (Xu et al., 2023).
 - GroundedSAM for text-conditioned detection and masks (Liu et al., 2023; Kirillov et al., 2023).
 - Provide **competitive baselines**
- Proposal upper bound estimates achievable performance ceiling
 - Best-overlap proposal per frame using SAM (Kirillov et al., 2023).
 - Highlights remaining headroom for improvement
 - Gap indicates **proposal limitations**

- VISOR-NVOS J and F: 42.4 (ViT-L/14)
 - +18.4 vs SAM+CLIP (Kirillov et al., 2023; Radford et al., 2021)
 - +2.9 vs GroundedSAM (Liu et al., 2023; Kirillov et al., 2023)
- VOST J_ins: 26.7
 - +7.5 vs SAM+CLIP (Kirillov et al., 2023; Radford et al., 2021)
 - +1.4 vs GroundedSAM (Liu et al., 2023; Kirillov et al., 2023)
- Same ViT-B/16 encoder comparisons
 - +14.1 (VISOR-NVOS J and F) over SAM+CLIP (Kirillov et al., 2023; Radford et al., 2021)
 - +6.2 (VOST J_ins) over SAM+CLIP (Kirillov et al., 2023; Radford et al., 2021)
- Gap to SAM upper bound remains
 - VISOR-NVOS J and F: 73.0 (SAM proposals)
 - VOST J_ins: 65.8 (Kirillov et al., 2023)
- Takeaway: consistent gains; proposal quality caps headroom

- Global video-narration objective yields substantial improvements
 - Boost over frozen CLIP encoders alone
 - Captures clip-level semantic alignment
 - Provides **robust supervision**
- Region-phrase objectives further enhance mask selection
 - Both region-to-phrase and phrase-to-region help
 - Combined objectives unlock additional gains
 - Direct supervision sharpens **pixel alignment**
- Adaptive temporal pooling outperforms naive aggregation
 - Resists occlusion and motion blur
 - Balances best-frame and multi-frame evidence
 - Improves **temporal robustness**



- Model segments targets amid clutter and occlusion
 - Resolves distractors in crowded scenes
 - Maintains mask quality under overlap
 - Visual results indicate **robust localization**
- Model tracks objects through large appearance changes
 - Sustains masks during manipulation
 - Handles deformation and partial visibility
 - Consistent masks show **shape invariance**
- Model disambiguates closely related phrases
 - Separates similar items by attributes
 - Handles packaging versus content distinctions
- Guidance: look for disambiguation and steady masks under occlusion



- Partial masks arise from ambiguous object extent
 - Large appliances segmented only in part
 - Unclear boundaries reduce IoU
 - Errors reflect **scale ambiguity**
- Plural references are grounded as single instances
 - Multiple targets merged or ignored
 - Phrase count not fully captured
- Future improvements focus on temporal and proposal quality
 - Leverage motion and consistency signals
 - Enhance proposal ranking and recall
 - Address attributes to **reduce ambiguity**
- Guidance: note partial masks on large objects and missed plurals

- Model transfers to YouCook2-BB with competitive box accuracy
 - Macro accuracy of 37.93 and micro of 44.96
 - Comparable to prior methods on YouCook2 [41,48,56,64]
 - Cross-domain performance shows **strong transfer**
- Point grounding substantially outperforms prior approaches
 - Point accuracy of 69.35 versus baselines
 - Large absolute gain over CoMMA† (Gao et al., 2023) and CoMMA (Gao et al., 2023)
 - Improvements validate **region alignment**
- Evaluation uses masks converted to boxes for fair comparison
 - Consistent protocol across methods
 - Directly reflects localization fidelity
 - Practical metric for **real-world use**

- Frame-wise inference underuses temporal continuity
 - Lacks explicit motion modeling
 - Temporal cues could reduce instability
 - Current design shows **temporal gaps**
- Domain focus on cooking constrains generalization
 - Performance drops in other activities
 - Broader training data is needed
 - Dataset diversity can **boost transfer**
- Proposal quality and plurality handling remain open challenges
 - Improve recall and mask fidelity
 - Explicit multi-instance grounding for plurals
 - Attribute reasoning reduces **misalignment**

- Weakly supervised NVOS enables phrase-level masks from narration alone
 - No reliance on spatial labels at training
 - Instance-aware segmentation in egocentric videos
 - Language-only pipeline is **scalable**
- Global-local learning delivers state-of-the-art zero-shot results
 - Strong gains on VISOR-NVOS and VOST
 - Clear advantages over stronger supervised baselines
 - Comprehensive **pixel grounding** benefits
- Trained models transfer to exocentric grounding tasks
 - Competitive box and point accuracy on YouCook2-BB
 - Indicates robust cross-domain alignment
 - Future work will **expand domains**

Thank You

Questions & Discussion